

CLEAN-1 through CLEAN-5: Summary of Changes

Revision 2b — GATE 2 Review

Manuscript: Spatial CRT

GATE 2 diff — C6 snapshot → CLEAN state

This document summarizes all visible changes made in the CLEAN-1 through CLEAN-5 pass. CLEAN-1 (comment deletion: 601 lines of commented-out draft text) is not shown here because all deleted content was invisible to readers (% prefixed). CLEAN-5 was a no-op (no reviewer-attribution language found in the manuscript body).

CLEAN-2: Remove Redundant $\rho = 0.01$ Sentences

Rule: The closing “ $\rho = 0.01$ increases MSE” generalizing sentence was kept only in §3.4 (2×4 grid section). The equivalent sentences in §3.5 and §3.6 were removed as redundant cross-grid repetition.

Change 2a — §3.5 (3×3 grid), end of main paragraph:

~~Similar to the 2×4 grid, the presence of spatial dependence ($\rho = 0.01$) generally increased MSE for both sampling methods across both spillover application types for the 3×3 grid.~~

→ *[sentence deleted]*

Change 2b — §3.6 (3×4 grid), opening of concluding paragraph:

~~Across all grid sizes, increasing ψ consistently raises MSE for both methods, and the presence of spatial dependence ($\rho = 0.01$) further inflates MSE relative to the $\rho = 0$ setting. The 3×4 results reinforce the broader conclusion: BSS reliably bounds worst-case estimation error under TrtNoSpill, while the TrtSpill regime with spatial dependence represents the primary scenario in which a practitioner should carefully weigh the risk-management benefits of BSS against its potential for amplified bias.~~

→ *[red sentence deleted; blue sentence retained]*

CLEAN-3: Remove Appendix D (Moran’s / Geary’s)

Rule: Appendix D (Moran’s I / Geary’s C formulas) was a supplementary appendix that added length without being needed in the results. The cross-reference in §2.6 was replaced with an inline `spdep` citation.

Change 3a — §2.6 (Spatial Dependence), last sentence:

~~Formal expressions for both statistics are provided in Appendix D.~~

Both can be computed in R via the **spdep** package [Bivand 2018].

Change 3b — Appendix D (entire section) deleted:

*[DELETED: “§Appendix D: Tests of Spatial Dependence: Moran’s I and Geary’s C”
—approx. 25 lines including Moran’s I formula, Geary’s C formula, and two R function references. Appendix E (Distance-Based Neighbor Construction) auto-renumbers to Appendix D.]*

CLEAN-4: Appendix A Notation Fixes

Rule: Raw weights renamed $w_{ij} \rightarrow \tilde{w}_{ij}$ to distinguish from row-normalized W_{ij} ; “subjects” corrected to “clusters”; row-normalization denominator corrected to $\sum_{j \neq i}$ (self-exclusion).

Change 4a — Indicator-based weight definition:

$$\underline{w_{ij}} \tilde{w}_{ij} = \begin{cases} 1, & \text{if subjects clusters } i \text{ and } j \text{ are Rook neighbors} \\ 0, & \text{otherwise} \end{cases}$$

Change 4b — Indicator-based row normalization:

$$W_{ij} = \frac{\underline{w_{ij}} \tilde{w}_{ij}}{\sum_{\underline{j} \neq i} \underline{w_{ij}} \tilde{w}_{ij}}$$

Change 4c — Distance-based weight definition:

$$\underline{w_{ij}} \tilde{w}_{ij} = \begin{cases} \frac{1}{d_{ij}}, & \text{if } 0 < d_{ij} < d_{\max} \\ 0, & \text{otherwise} \end{cases}$$

where d_{ij} is the distance between subjects clusters i and j .

Change 4d — Distance-based row normalization:

$$W_{ij} = \frac{\underline{1/d_{ij}} \tilde{w}_{ij}}{\sum_{\underline{j} \neq i} \underline{1/d_{ij}} \tilde{w}_{ij}}$$

CLEAN-5: Reviewer Attribution Audit

Result: Verified clean. Grep for “Reviewer” in manuscript body found only one hit (line 707): “The authors thank the Editor and both Reviewers for their thorough and constructive feedback. . .” This is in the Declarations / Acknowledgements section — appropriate academic acknowledgment, not manuscript-body attribution. No edits required.

Page Count

State	Pages	Lines
V2b after C6 (baseline)	48	1472
After CLEAN-1 (comment deletion)	—	871
After CLEAN-2–5 (current)	42	844

Six page reduction (48→42) primarily from removal of Appendix D and elimination of the overly wide double-spaced table formatting (already applied in C4).